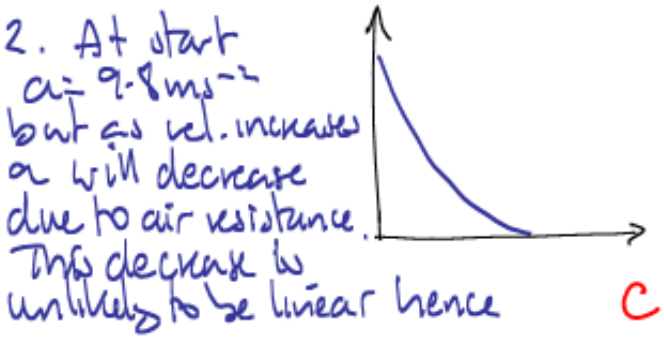
P1 May 2009 TZ2

1.

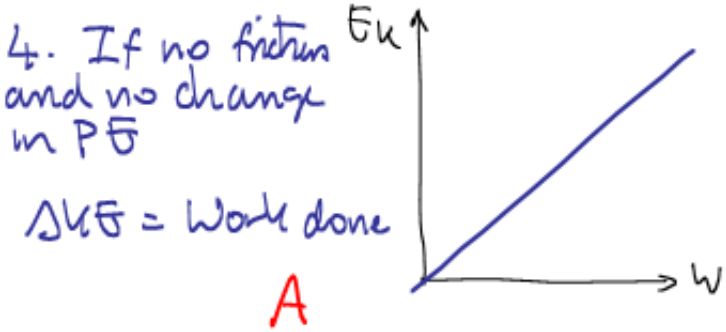
1. Some things you just have to remember, however it's a good guess that it's going to be the biggest one

Universe = V. big D

2.



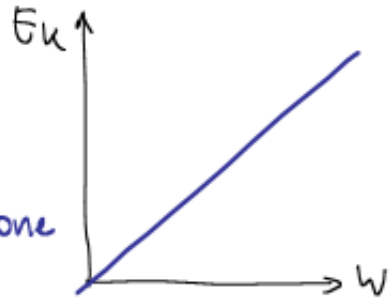
3.



4.

4. If no friction and no change in PE

ΔKE = Work done



A

5.

5.



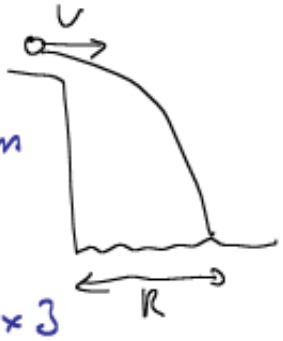
Only one force acts on the moon - the gravitational force.
- This force will provide the centripetal force
- They are the same thing -

A

6.

6. Time of flight depends upon the vertical motion - unchanged

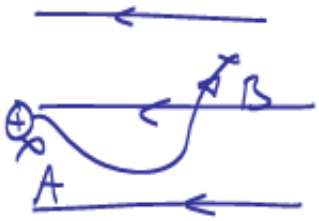
Range depends on horizontal velocity × 3



C

7.

7.

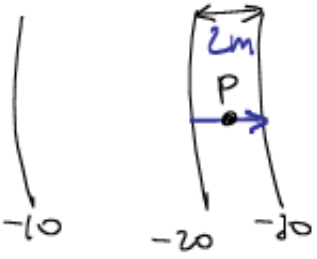


Work done depends on the size of charge and the difference in potential $W = \Delta V q$

So not path or speed B

8.

8.



field strength $= \frac{\Delta V}{\Delta d}$

$= \frac{10}{2} = 5 N kg^{-1}$

direction is from high → low

$-20 \rightarrow -30$ B

9.

9. Temperature is a measure of the average KE of gas molecules C

10.

10. Ideal gas equation $PV = nRT$

const m ⇒ const no. of moles

but $\rho = \frac{m}{V} \Rightarrow P \propto \rho T$

$P \propto T/V$ B

11.

11. A – No it's not possible
B – Local entropy can decrease but entropy of universe must increase.
C – Yes
D – entropy in a closed system will increase

Have to read this one carefully C

12.

12. The central point is when the resultant force = 0 ⇒ equilibrium.
B

13.

13. F α distance from a fixed point
A

14.

14. Refractive index = $\frac{\text{velocity in 1}}{\text{velocity in 2}}$

$= \frac{f\lambda_1}{f\lambda_2} = \frac{\lambda_1}{\lambda_2}$

measured from the diag = $\frac{1}{1.5}$

$n = 0.67$

B

15.

15. In a standing wave:

- Energy not transferred ✓
- $I/S \propto A^2$
- between nodes strings in in phase
- Amplitude varies with position not time

A

16.

16. frequency is lower since the waves are spread out
⇒ greater λ



D

17.

17. high energy e⁻s have short wavelength so reducing diffraction but increasing resolving power.

C

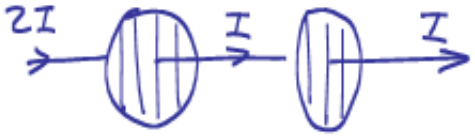
18.

18. Brewster angle $\theta = \tan^{-1} n$
It's in the data book!

A

19.

19.



1st polariser reduces Intensity from 2I to I, the second one does nothing.

If either rotated by angle θ the intensity will become $I\cos^2\theta$

If Intensity $= \frac{I}{2} = I\cos^2\theta$

$\frac{1}{2} = \cos^2\theta \quad \cos\theta = \frac{1}{\sqrt{2}}$

$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$ C

20.

20. eV - unit of Energy.

The amount of energy gained by an electron accelerated by a pd of 1V

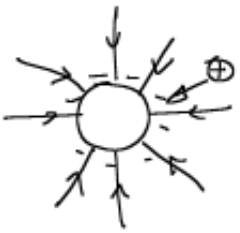
C

21.

21. Ideal ammeter has zero R so doesn't reduce current
Ideal voltmeter has ∞ R so does not draw current C

22.

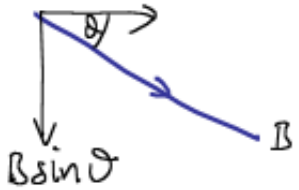
22. field is radial and towards the −ve sphere



B

23.

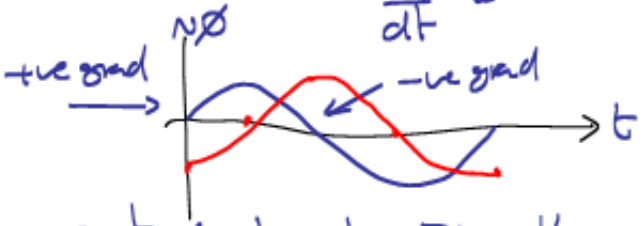
23. Take comp of B ⊥ to Area



$\pi r^2 \times B\sin\theta$ B

24.

24. $\mathcal{E} = -\frac{d N\phi}{dt}$ −ve sign ⇒ when $\frac{dN\phi}{dt}$ is +ve $\mathcal{E}$ is −ve



out of phase by π/2 − ¼ cycle B

25.

25. $\mathcal{E} = -\frac{dN\phi}{dt}$ so if frequency is doubled rate of change is doubled ⇒ $\mathcal{E}$ doubled

⇒ $2V_0$ C

26.

26. photon has $E = hf$
momentum $= \frac{h}{\lambda}$

so has both energy and momentum.

This is strange since a photon has no mass but it can be explained by relativity. A

27.

27. de Broglie hypothesis applies to anything with momentum
⇒ all particles D

28.

28. If particle is modeled by wave function ψ then the probability of it being at a given position is found from ψ^2

D

29.

29. α - most ionizing
β
γ - least ionizing

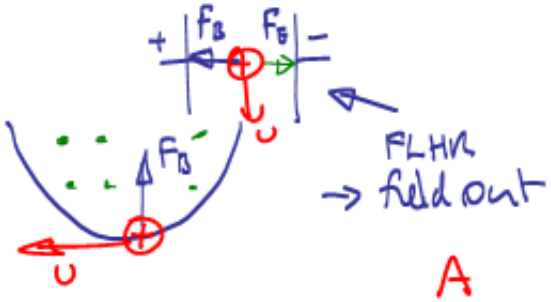
So order in increasing ionizing power is $\gamma \beta \alpha$

D

I was caught out on this one. Read carefully.

30.

30.



A

31.

31. - If fuel rods removed reaction slows down
- If control rods pushed in reaction slows down
- If moderator removed reaction slows down.

=> All 3 maintain the rate of fusion A

Answer given as A or C since it is the control rods that are actually used to control the reaction.

32.

32. $^{1}_{0}n + {}^{198}_{80}Hg \rightarrow {}^{197}_{79}Au + {}^{2}_{1}H$

is a transmutation, I'm not sure how you should know that it is artificial but it is.

D

33.

33. Binding energy is the energy released when a nucleus is made or required to split the nucleus into its parts

A

34.

34. Power per unit length = $\frac{1}{2}A^2\rho s v$

from data booklet

$= P_d A^2$

The energy in a wave is KE + PE

D

35.

35. $P = \sigma T^4$

so if 20× temp then 20×20×20×20× power

$= 160{,}000P$

D

36.

36. All though some think otherwise most scientists agree its burning of fossil fuels

C

37.

37. Reducing snow reduces the reflected light ⇒ reduces albedo

increasing E absorption

B

38.

38. When the light hits an edge → 1, this is when the interference is destructive

B

39.

39. Electrons are liberate by the photoelectric effect

A

40.

40. $M = \frac{\text{image size}}{\text{object size}} = \frac{0.03 \times 10^{-3}}{30}$

$= 10^{-6}$

D

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